

Phishing Attacks in Social Media

Abstract—Phishing attacks on social media are one of the most critical cybersecurity threats that target sensitive information and users' identities. Cybercriminals exploit the amount of information that social media users share on these platforms through several techniques, including deceptive messages and malicious links that pretend to be legitimate. This study reviews existing literature to analyze social media phishing techniques and identifies gaps in real-time detection of it, and based on these findings, it proposes a framework for phishing detection and prevention strategies on social media.

Keywords—Phishing attacks, social media security, cybercrime, user awareness, machine learning, detection framework, social engineering.

I. INTRODUCTION

The widespread use of social media platforms introduced significant cybersecurity challenges. And one of these consequential attacks is phishing, which occurs when hackers exploit the vast amount of personal and sensitive information shared by users. Unlike traditional email phishing, social media phishing utilizes platform-specific features such as likes and shares and users to get the victim's trust and increase the likelihood of user interaction.

The consequences of successful attacks include identity theft, financial loss, and compromise of organizational data, which demonstrate the need for detection and prevention strategies.

Although people are awareness of phishing threats, Increase significantly, research has primarily focused on email phishing, leaving social media-specific attacks unresearched.

The primary objective of this study is to examine phishing attacks in social media platforms, evaluate existing detection mechanisms, and propose a framework to address identified vulnerabilities.

II. LITERATURE REVIEW

Evaluation of the Effectiveness of Cyber Awareness Training on Phishing Scam on Social Media. This research discusses the increasing problem of phishing scams on social media networks such as Instagram [1]. In such instances, the attackers trick users into giving away sensitive data such as passwords or personal information via bogus links or pretending to be someone they know. This is largely based on social media where information is easily and quickly shared and users act without thinking.

The research found that training people on cybersecurity is useful. It enhances digital literacy levels and assists users in spotting phishing attempts. However, the problem is that knowledge is lost over time. Furthermore, many users are just not that engaged so awareness alone is not enough to prevent the attacks. This is where machine learning comes into play. Phishing detection is supported by several techniques, including URL analysis, behavioral analysis, and natural language processing (NLP), each based on different mechanisms. NLP is very good at identifying red-flag language but can be tricked by more cleverly worded messages.

URL analysis works well for identifying malicious links that have been present in the system for some time but struggles to detect new links. Behavioral analysis can identify abnormal user activities but also can generate false positives. Studies have shown that the best approach is a combination of several methodologies because each of them has its own limitations. These models improve accuracy and cover more angles of an attack. So, the bottom line is the strongest defense against phishing on social media isn't just educating users or just using AI; it's a smart mix of both.

[2] Phishing exploits human cognition through authority, urgency, and familiarity. Even skilled users are targets of personalized attacks. Awareness is not enough, as security awareness training only improves detection by 20-30%, and has a rapid decay. AI-generated phishing messages are now as effective as spear-phishing, but at a 50% lower cost. Attackers scour LinkedIn and Instagram to create hyper-personalized lures. Voice cloning has also transformed vishing; a cloned CEO's voice asking for a wire transfer duped 65% of participants.

Phishing has moved beyond email to include SMS, QR codes, and social media. Multi-stage attacks, where trust is built before credentials are requested, have a 45% success rate — nearly double that of single-shot phishing. Traditional spam filters have failed against zero-hour URLs (average lifespan: under 15 minutes). Standard MFA is bypassable via fatigue attacks, SIM swapping, and adversary-in-the-middle proxies—prompting NCSC and CISA to recommend FIDO2/passkeys instead.

We found that what works is a layered defense combining FIDO2/passkeys (zero successful phishing in controlled trials), AI-native behavioral detection (>98% accuracy), and out-of-band verification habits, representing the current best practice. In conclusion, AI has made phishing more scalable and harder to detect. Traditional training and MFA are no

longer enough. The solution resides in phishing-resistant, zero-trust architectures.

[3] The literature review summarizes the main findings of the extensive study by Tripathy (2024) on cybercrime in India within the global digital ecosystem. The study reveals a significant rise in cybercrime incidents in India, from 9,622 in 2014 to over 77,000 as of August 2024, attributed to increased internet penetration, digital payments and vulnerabilities in the financial, healthcare and government sectors. The study provides a detailed taxonomy of crimes, including financial scams, phishing, identity theft, ransomware and emerging cybercrimes such as deepfakes and dark web activities. The rise in arrests from 5,752 to 36,235 further illustrates the increasing nature of cyber threats and the challenges faced by law enforcement.

[4] This literature review considers the doctrinal review of Nigeria's Cybercrimes (Prohibition, Prevention, etc.) (Amendment) Act 2024 by Onwuzuruoha and Akinola (2024), which discusses the legal protections for social media use considering Nigeria's changing landscape of cybercrimes. The review first introduces key ideas surrounding cyberspace, cybercrime and social media, before providing a detailed list of the broad scope of crimes in the Amendment Act, including crimes against critical national infrastructure, unauthorized access to computer systems, registration requirements for cybercafes, interference with systems, interception of electronic communications, computer-related forgery and fraud, and cyber terrorism, and then also poses the legislation within Nigeria's wider legal landscape, including the 1999 Constitution's protections of freedom of expression, the Nigerian Data Protection Act 2023 and the Advertising Regulatory Council of Nigeria Act 2022.

Protection Act 2023 and the Advertising Regulatory Council of Nigeria Act 2022. Through a comparative review of the cybercrime regimes in the United Kingdom (namely the Online Safety Act 2023), Canada (including the proposed Online Harms Act) and South Africa (Cybercrimes Act 19 of 2020 and POPIA), the authors identify important lessons for Nigeria, especially the importance of making social media platforms accountable for harmful content rather than solely criminalizing individual offences.

The review ends with the note that the 2024 Amendment Act is an important step forward but that more steps need to be taken to effectively prevent cybercrimes including the introduction of cybersecurity education in school curricula, major investments in cybersecurity infrastructure, better law enforcement training and public-private partnerships with technology companies to combat emerging digital threats. Vanna et al. provides a comprehensive and critical review of advanced hybrid techniques for the detection of mobile phishing attacks.

The authors point out that the threats against mobile devices are becoming more complex, being delivered through malware applications, SMS, social media, and QR codes. Current detection methods are not sufficient to fight against the evolving threats, and hybrid techniques are necessary, combining machine learning, deep learning, and heuristic rules. The authors perform a systematic review of peer-reviewed research from 2018 to 2024 from IEEE, Springer, and ScienceDirect databases and plot the evolution of

phishing detection from traditional methods to advanced hybrid models for mobile environments.

The survey divides the phishing detection techniques into supervised learning techniques such as logistic regression, decision trees, random forest, support vector machines, and naive Bayes, which achieve accuracies from 90% to 99.45% on different datasets, and deep learning techniques such as convolutional neural networks (CNN), recurrent neural networks (RNN), and BERT, which reach up to 99.4% accuracy. The survey is focused on mobile-specific phishing attacks in the Android and iOS ecosystem to cover attack vectors such as SMS phishing (smishing), voice phishing (vishing), QR code phishing (quishing), fake mobile apps, and social media deception. The research demonstrated that hybrid models that combine different techniques, such as ensemble learning (Random Forest with XGBoost), deep learning hybrids (CNN with BiLSTM), and edge computing with federated learning, consistently outperform single-method approaches. The detection accuracies in mobile security scenarios ranged from 97% to 99.82%.

A significant contribution of this work is its examination of emerging technologies reshaping phishing detection. Authors propose the use of federated learning for privacy-preserving distributed detection, blockchain-based secure DNS verification and real-time domain banning through smart contract, Explainable Artificial Intelligence (XAI) with SHAP and LIME for model explainability and user trust, and quantum machine learning as a new frontier for processing large scale data with better speed. The study also talks about the implementation challenges such as computational complexity, scale requirements, data privacy issues under regulations such as GDPR and CCPA, and mobile-specific constraints such as small screen interfaces that restrict URL visibility and user awareness of phishing indicators.

Authors conclude that future directions should focus on privacy-preserving techniques, cross-domain detection capabilities, real-time protection mechanisms, and the integration of XAI and quantum computing to improve both explainability and cybersecurity effectiveness in mobile phishing detection systems.

[6] The present study aims to contribute to the literature on the topic of cybersecurity risks and artificial intelligence (AI) in the ever-evolving environment of online social networks (OSN) and technology. The main objectives are to classify cyberattacks such as malware, phishing/spam and network intrusion detection; to identify effective algorithms for preventing cyber threats; to review relevant literature from 2019 to 2020; and to detect suspicious behavior related to malware using machine learning algorithms. The study proposes a novel framework that proposes specific machine learning algorithms for each type of cyber threat, thereby improving cybersecurity knowledge and response capabilities. This makes the research useful for examining the impact of cybersecurity on smart cities.

The study's conclusions center on how AI and machine learning algorithms linked to cybersecurity are reviewed in the literature, how cyberattacks are classified, and how an inventive framework for identifying and reducing risks is proposed. This makes the research useful to research the implications of cybersecurity for smart cities.

- **Practical implications**

The practical implications of this research are noteworthy, particularly in the realms of technology, AI, machine learning, and innovation. The utilization of the NVivo technique enhances decision-making in uncertain situations, making the study's results more reliable. The findings showcase the applicability of tools in analyzing malicious cyberattacks to address issues related to social media attacks, emphasizing their practical utility. The study's relevance is further highlighted by a real-world example, where a Kuwaiti public sector fell victim to a malware attack, underlining the importance of cybersecurity measures aligned with the New Kuwait 2035 strategic development plan. The innovative framework presented in the research guides the selection of algorithms for detecting specific malicious attacks, offering practical insights for securing information technology (IT) infrastructure in Kuwait.

- **Social implications**

The rapid digitization in Kuwait, accelerated by the COVID-19 pandemic, underscores the pivotal role of technology in government services. Ma'murov et al. (2023) emphasize the significance of digitization, particularly in accessing and verifying COVID-19 information.

The call for a dedicated digital library for preserving pandemic-related material aligns with the evolving digital landscape. Cybersecurity emerges as a critical concern in Kuwait and the Gulf Cooperation Council (GCC), necessitating transnational cooperation (Nasser Alshabib and Tiago Martins, 2022). In the local context, the inefficiency of information security systems and low awareness among government employees pose cybersecurity challenges (Abdulkareem et al., 2014). Social media played a significant role during the pandemic, yet the need for cybersecurity in this area remains underexplored (Ma'murov et al., 2023; Safi et al., 2023).

[7] Madlenak and Hubocan, in their 2025 study, in their 2025 study look into how people can spot content made by AI. This is a deal because of something they call "Phishing 2.0 " which is the next level of phishing attacks that use AI. Traditional phishing usually has mistakes like grammar errors and weird requests. Ai-driven attacks use tools like GPT-4 to make content that is very personalized has no errors and is very convincing across different ways of communicating.

The authors compare phishing methods like bulk phishing, spear phishing, and whaling to their new AI versions. They show how AI makes it hard to spot phishing by getting rid of the mistakes people used to rely on to detect them. The study had 21 people from infrastructure security, transport security, crisis management and security management fill out a 12-question survey. They had to tell if content was made by AI or a human across four areas: portraits, company logos, SMS messages, and emails. The results show that these professionals got it right 57% of the time. They were better at spotting content like human portraits at 66.7% and company logos at 58.7%. But they struggled with text-based content: SMS messages were correct 50.8% of the time and emails 52.4%. This means that while AI images might still have some giveaways, AI text is getting really good and hard to tell from human-made text.

The study's findings are important for cybersecurity for areas like transportation, energy and healthcare that are more

vulnerable to AI phishing attacks. The authors think that old training methods focusing on grammar errors might not work against AI phishing attacks.

They suggest research, bigger groups of people, and new ways to train people to spot AI content. They conclude that as AI gets better, it's crucial to educate people, about AI threats to stay online.

[8] This study aims to model and analyze the strategic interactions between phishing call center scammers and authorities using game theory. By identifying equilibrium stability conditions, it provides insights into optimizing cybersecurity defense strategies.

- **Methodology**

A game-theoretic model was developed to represent the attacker (phishing call center scammers) – defender (authorities) interactions. The model included key parameters such as costs, rewards, penalties, and success rates. The trace-determinant method was applied to analyze the stability of the system, especially the four pure strategy equilibrium points, by deriving inequalities from the eigenvalues of the Jacobian matrix. Stability conditions were analyzed under general and assumptions to reflect realistic cybersecurity scenarios.

- **Findings**

The research found that individual-target defense is more effective for low-cost attacks, while investment in overarching defense is more effective for high-cost attacks. The stability analysis has identified critical factors affecting the long-run strategic equilibrium between attackers and defenders, including costs, arrest and defense rates, and rates of successful attack.

[9] Klutsch et al. (2025) consider the susceptibility of young adults to social media phishing from a developmental psychology perspective, emphasizing how a relationship with the phisher and developmental opportunities offered in messages affect vulnerability. In an experimental vignette study involving 500 young adults (aged 16–29), participants received Instagram messages from followers or non-followers offering social, job, or partner opportunities, corresponding to key developmental tasks, i.e., belonging to a peer group, entering professional life, and being in a committed relationship.

The findings show two key takeaways. First, followers' messages significantly increased susceptibility compared to non-followers. Second, social opportunities, messages inviting participation in events or activities, generated the highest susceptibility, with almost two-thirds of participants responding to such messages, and this effect was magnified when social opportunities were from followers. Job opportunities generated higher susceptibility than partner opportunities, especially from non-followers. The results suggest that young adults are more likely to process messages heuristically using follower status as a credibility cue, rather than systematically processing message content. Phishing awareness campaigns should leverage social media platforms, increase the consequences of compromised trusted accounts, and tailor interventions for the young on their most used platforms.

[10] Yasin et al. (2024) studied the effectiveness of serious game-based learning to build resilience against phishing and spear-phishing attacks and highlighted that humans are the

weakest link in information security. The authors developed “PhishDefend Quest”, a hybrid serious game that employs QR codes to access real-time data, and conducted an empirical study with 97 participants using surveys, observations, discussions, and pre-post URL assessments. The game design is based on information security, game design, and human-computer interaction principles and is guided by Bloom’s Taxonomy, the MITRE ATT&CK framework, and social engineering ontology, following a four-stage process: target selection, information gathering, drafting phishing emails, and dissemination. The empirical findings show good learning results.

Results from the pre-post URL survey shown significant improvement in participants’ ability to identify phishing URLs with accuracy levels rising from 27.8% to 80% for certain URL types. Regression analysis shows a significant relationship between perceived threat, avoidance motivation, and avoidance behavior, supporting the proposed paths. Qualitative observations suggest that participants actively took the role of the attacker to better understand how phishers exploit the overexposure of online information. The authors conclude that game-based learning is an effective method to educate participants about the threats of phishing and the hazards of overexposure of online information and suggest future research with larger sample sizes, longitudinal research and expansion to other platforms.

[11] Bossetta (2018) studies the use of social media by state-sponsored actors to implement spear phishing and cyber espionage campaigns against Western democracies and argues that social media platforms are particularly attractive for sophisticated cyberattacks because of their high levels of trust, habitual use and personal data. The author develops a five-stage theoretical model: Collect (discovery and reconnaissance of target data), Construct (building of fake profiles with inflated metrics), Contact (initiation of interactions through messages or adverts), Compromise (deployment of malware or hijacking of accounts), and Contagion (viral dissemination through compromised accounts).

Empirical examples include Russian Fancy Bear operations against the Macron campaign, Chinese intelligence creating fake LinkedIn profiles and Iranian operatives inventing personas such as “Mia Ash.” Bossetta notes that spear phishing attacks on social media increased by 500 percent in 2016, with one Russian operation realizing a 70 percent click-through rate on malicious links sent to more than 10,000 U.S. Defense Department employees. The author concludes that democracy is under attack when its fundamental tenet—trust—is exploited for illiberal pursuits.

[12] Parker and Flowerday did a study in 2020 to figure out why some people who use social media are more likely to fall for phishing scams than others. They looked at Facebook, Instagram and Twitter because phishing scams are moving from email to these platforms.

They read 25 articles from Emerald Insight, ScienceDirect and Google Scholar. Used a special tool called MAXQDA to analyze them. They found out that there are five things that make people more susceptible to phishing scams: how they process information, their habits, who they are what they know about technology and their personality traits. People who do not think much about what they see online and just go with what looks okay are more likely to fall for phishing

scams. If people use media a lot and do things without thinking like clicking on links or liking posts they are more likely to get scammed.

The study showed that young adults people from 18 to 25 years old and women are more likely to fall for phishing scams. Young people do not know much about phishing and are not as careful with their money. Women are not as good with technology. They can spot scams if they pay attention. If people know a lot about computers. How to stay safe online they are less likely to fall for phishing scams. Some personality traits also make people more likely to fall for scams. People who are outgoing open, to things and friendly are more likely to get scammed but people who are careful and responsible are less likely to fall for phishing scams.

[13] Alwanain (2020) looked into how phishing awareness training works for older adults on social media. This is a deal because older adults are often targeted by phishers. They have a lot of money are less likely to report fraud and don't know much about risks. In 2019 people over 60 had 68,013 complaints. Lost \$835 million to scams according to the IC3. Phishers often target popular platforms like WhatsApp, Facebook and Netflix. WhatsApp is a way for scammers to trick people. The study had 20 participants aged 60 to 75. They were split into two groups. In the part of the study the treatment group got a phishing message with a fake link about COVID-19.

The control group got the message but with a real link. The treatment group got some education about phishing. Two months later both groups got another message. The results showed that 87.5% of the control group clicked the link at first compared to 72.7% of the treatment group. After two months the treatment group got better at spotting phishing messages. 54.5% Of them clicked the link down from 72.7%. The control group also got a bit better dropping from 87.5%, to 62.5%. The study found that the training really helped the treatment group. This means that phishing awareness training can work for adults. The study also found that 60% of all participants clicked the link on the day. This shows that many older adults struggle to identify phishing messages.

[14] Alam and El-Khatib (2016) explore the relationship between social media presence and susceptibility to spear phishing, recognizing that personalized attacks achieve far higher success rates—70 percent open rates versus 3 percent for generic phishing—due to attackers' ability to gather personal information through passive social media reconnaissance. The authors synthesize prior research showing that users with certain personality traits (extraversion and openness) are more susceptible to phishing and that these traits can be accurately predicted through social media activity.

Building on this, they propose a novel mitigation concept: rather than detecting spear phishing emails, individuals should instead manage the information available about them online. By reducing the accuracy of an attacker’s profile—for example, by diluting strong interests with mixed content—the effectiveness of personalized attacks can be reduced. To operationalize this, the authors present a modular architecture that includes a Flow Control Engine, platform-specific data preparation engines, and data analysis engines for topic analysis. Preliminary testing confirms that a significant amount of identifiable information is publicly available, indicating that awareness of online exposure and strategic

dilution of interests are viable approaches to reducing phishing vulnerability.

[15] Silic and Back (2016) studied the risks of phishing in the context of organizational social networking using a multimethod study consisting of a field experiment in a Fortune 500 company, interviews with ten employees and interviews with eleven chief information security officers (CISOs). The field experiment consisted of creating a fake LinkedIn profile impersonating an employee who was able to access a private company group without verification. A fake webpage link was posted to the group, which was clicked on 180 times by employees and 122 of them (68%) provided personal information, which shows that contextual elements and the “liking” influence technique make employees highly vulnerable to SNS phishing, especially females and younger workers. The control group using student profiles achieved only 9% success.

Four main results emerged from the study: employees are susceptible to deception on SNSs where contextual elements act as psychological triggers; organizations have no measures to control the security threats posed by SNSs; companies need stronger information security policies that address SNS identification and authentication; and SNSs have become critical security holes, especially considering the “multiplication effect” of single sign-on technologies that allow one compromised credential to expose multiple applications. The authors conclude that existing security policies and training are inadequate for SNS realities, and recommend updated policies, stronger authentication, and training that addresses SNS-specific influence techniques.

[16] Badescu in the year 2024 did a detailed study of phishing threats. This study looked at things from the users point of view. It used a review of what other people have written about this topic and it also included a survey of 47 students who are studying to be cybersecurity masters. The study found out that when people know about phishing they are better at stopping it. Even people who know a lot about computers can still get tricked by phishing. In fact one study phishing attacks, but it's still important for users to be aware found that the more people know about phishing the likely they are to fall for it.

The person who did the study came up with a way to score types of phishing. They looked at 16 kinds of phishing. Gave them scores. The types that got the scores were spear phishing, whaling and business email compromise. These types of phishing got scores of 90 to 100 because they are very personalized and can cause a lot of damage. On the hand regular email phishing got a lower score of 60 to 70 even though it can reach a lot of people. There are some phishing threats that are very scary. For example some emails are written by intelligence and do not have any spelling mistakes.

There are also voices that can sound like real people like what happened in a case where someone pretended to be a CEO in the UK. And there are phishing kits that can get around security measures that require steps to log in.

The survey found that students who are studying cybersecurity are very aware of phishing threats. They are very careful when it comes to who's sending them emails and what links they click on. They are not as good at spotting generic greetings or grammar mistakes in emails. The study suggests that we should teach kids about cybersecurity in

school use videos to train people and make awareness campaigns for older adults, on TV.

The person who did the study thinks that education is the way to stop phishing. They say that education is the one thing that cybercriminals cannot ruin and it should be used along with ways to stop phishing like using artificial intelligence to filter emails and analyzing the way people write.

[17] This research presents a full overview of social engineering attacks and especially phishing, which is one of the most common and damaging cyber threats. Social engineering is the act of influencing people's actions by exploiting their trust, curiosity, and disinterest. It is not about exploiting computer system vulnerabilities. It explains how attackers do reconnaissance on their targets, gain their trust, exploit that trust to attack and all that without being caught. They use psychological tricks to get what they want out of people and it's a big problem because it can lead to serious security breaches. When we know how social engineering attacks work, we can do a better job of protecting ourselves and our systems against these types of threats.

The study also covers the different ways to detect phishing, including based on rules methods, lists of known bad sites, and machine learning. Every way has its pros and cons points, but some, like machine learning, rules that are unclear, are better at catching phishing attacks.

Social engineering is one of the biggest threats to the cybersecurity field. It's when someone tries to trick the victim into doing something that puts security at risk. The attack is not on the technology directory but on the people that use it. There is no shortage of tech solutions to detect and stop of the risks and know how to protect themselves, the study adds. So to reduce the risk of these types of attacks, we need a combination of technical defenses but also make sure users are educated and aware of what to look out for.

[18] This research identifies phishing attacks as a major form of cybercrime which is aimed at acquiring sensitive user information such as passwords, banking details, and personal information. Phishing is a kind of social engineering attack in which attackers pretend to be trusted entities through e-mails, messages or websites to deceive users to reveal confidential information. The research elaborates the step-by-step procedure of phishing starting from the sending of fake communication and then collecting user credentials and exploiting them.

These include deceptive phishing, spear phishing, clone phishing, whaling, link manipulation, and voice phishing. They vary depending on the targets, and some target individuals, but others focus on high-end users or organizations. The different classifications show the continuous transformation of the attacks depending on the environment and user's behavior.

Also, this research talks about a number of prevention techniques, like avoiding suspicious emails, checking if connections are secure and encrypted, avoiding clicking on suspicious links, and having robust organizational security policies. Also, it argues that training and awareness reduces the success rate of phishing attacks.

In addition, the paper revisits the detection mechanisms that include the use of private DNS services, detection lists acquired through browser phishing, link-checking tools, and relying on users' judgment of suspicious sites. Using these

approaches together, phishing websites can be prevented. Grounded on human vulnerability rather than technical vulnerabilities, the study has come into the conclusion that phishing attacks remain a great threat. Thus, combining user awareness and security practices, and detection tools are key in minimizing risks associated with phishing attacks.

[19] Phishing attacks have been covered in a wide range of social engineering contexts in this study, with a focus on how cybercrime has become a more significant danger in recent years, particularly during the COVID-19 epidemic. Social engineering is thus a human-centered attack that preys on the users' behavior and trust, as well as their lack of awareness about the dangers, rather than on the vulnerabilities of the system. Phishing is identified as the most common and effective mode of attacks by attackers in obtaining sensitive information, including user login credentials, financial records, and personal information.

The paper discusses a range of social engineering attack options in the form of phishing, tailgating, pretexting, baiting, and impersonation (ibid), with each method pinpointing the exploitation of human vulnerabilities. Phishing is discussed in detail to demonstrate how attackers use fake emails, fraudulent websites, and unreliable links to convince a user to give away his confidential information. There are several classes of phishing attacks, depending on the way the attacks are propagated, with email phishing, vishing, smishing, spear phishing, and whaling being the main types of phishing propagation strategies adopted by cybercriminals and scammers and usually after considering the target and the medium involved.

In addition, the research includes an analysis of the impact of the COVID-19 pandemic on phishing activities showing a significant rise in the number of cyberattacks attributed to the move towards remote working and the increase in internet use. Statistical data presented in the study shows a sharp rise in phishing websites, emails, and targeted industries during this period, thus showing how attackers leverage global events to improve their attacks' success rates.

The study also investigates several phishing detection and prevention techniques, including machine learning techniques, blacklisting and whitelisting, URL similarity, web content analysis, and hybrid approaches. The study also considers various phishing detection and prevention techniques, such as machine learning approaches, blacklist and whitelist systems, URL similarity, web content based and hybrid methods, as well as the effectiveness of the different techniques, emphasizing that technical solutions alone are not enough in preventing phishing attacks (ibid). Instead, they necessitate the combination of technical and user-level interventions.

In addition, the paper demonstrates the importance of user behavior and psychological attributes regarding the susceptibility of phishing. Phishing Vulnerability Factors: Curiosity, Fear, Greed, and Trust The study also discusses personality traits and how such traits govern the users' likeliness to become victims of phishing attacks.

Lastly, the study finds that phishing attacks remain a big cybersecurity challenge since phishing attacks largely target human behavioral aspect. Therefore, more effective prevention needs to employ a balanced approach, gathering

technological solutions with user awareness, education, and strict organizational policies to against the risk of social engineering attacks.

[20] This research paper explores how social engineering and phishing have changed over the years, as technology has progressed. Social engineering is defined as a type of attack that depends on exploiting the human factor, such as the recipient's psychological vulnerabilities (for example; trust, fear or curiosity), instead of only using technical weaknesses. The research notes that phishing continues to be one of the major forms of social engineering and that it is often used to capture personally identifiable information (PII) like usernames, passwords, and banking account numbers.

In order to provide historical context to the topic of social engineering, the author describes how social engineering started with early telephone schemes of deceit and developed into the vastly more complex schemes of today that utilize advanced technologies including but not limited to artificial intelligence, deepfake technology, and automation.

Finally, the author approves the change from mass phishing to more refined, targeted attacks like spear phishing or business email compromise, where a specific person or organization is targeted by the attacker.

The research identifies and categorizes five types of social engineering attacks phishing, vishing, pretexting, baiting, and impersonation, and discusses how attackers use various manipulative techniques to gain the victim's confidential information. It also examines the underlying psychological principles that attackers use to manipulate how their victims think and make decisions, thereby increasing the likelihood of success with their scheme.

The research has looked at new ways to commit phishing attacks. Such attacks also use different types of technology (such as social networking and messaging services) as well as using different types of accounts (like fraudulent accounts) in order to attempt to entice users.

It has been found that the way an attacker approaches a phishing campaign may vary based on what is happening in the world as well as technological advancements that may have occurred during the last year.

The research identified different methods to defend against social engineering attacks, including the importance of using both technical measures along with awareness and training of users. It suggests forming a defence against phishing that uses multiple layers of defence, including artificial intelligence, analysing how people behave, and continually educating employees on best practices with cybersecurity.

The conclusion made by this report was that phishing will continue to adapt based on current events and technology, and therefore, it remains very important for both organisations and individuals to have a comprehensive defence against both types of vulnerabilities in the workplace.

[21] With rapid growth of phishing attacks, they are becoming a real risk to all users of the internet. Cybercrime continues to occur in large numbers on the global scale; phishing attacks are one of the fastest-growing forms of cybercrime and can pose serious risks to both individuals and businesses in an online-based environment. Phishing is defined as an attempt to fraudulently obtain sensitive

information, such as usernames, passwords, and banking information from legitimate users by masquerading as an identifiable entity (e.g., through e-mail, text message, or a website), while online.

This paper also discusses the evolution of phishing attacks over the last several decades. Phishing began with simple email scams in the 1990s, and as the internet has continued to expand globally, phishing attacks have continued to evolve into more complex and targeted attacks. The evolution of phishing also includes new modern phishing techniques, such as spear phishing, whaling, smishing, and vishing, and that one of the primary factors for the success of these various forms of phishing attacks is the attacker's use of social engineering and personalized information from targeted victims.

The study identifies emerging trends in phishing attacks, including the use of social media to facilitate more advanced and larger-scale attacks, as well as the increased use of artificial intelligence to conduct phishing attacks. Attacks are now being conducted with multiple vectors and utilizing a combination of e-mail, texting, and social media platforms and applications, thereby allowing attackers to successfully target their intended victims.

The analysis of the research revealed that phishing attacks have direct and indirect effects on those affected such as; financial loss, data breach, lost reputation/identity and psychological impacts. These represent very real consequences of phishing and their impact on both people and businesses.

The research also considers preventative measures to combat phishing attacks. Examples of these include; user awareness (education), antivirus software, two factor (2FA) or multifactor authentication (MFA), encryption and e-mail filtering systems. User awareness (education) appears to be one of the most important factors in successfully preventing phishing attacks due to human error, as opposed to technical deficiencies.

Accordingly, based upon the findings of the research, it is evident that the complexity and frequency of phishing attacks continue to rise. There is a need for a multi-layered approach to prevent phishing attacks. That's why organizations need to employ a combination of technology and training as well as strong organizational philosophies to help reduce cybersecurity risks.

[22] The paper deals with phishing attacks as one of the major cybersecurity threats that employs social engineering and technical techniques to deceive users and disclose sensitive data. The paper describes phishing as a hybrid attack in which hackers utilize different electronic communication techniques like emails, messages, and malicious links to guide users to counterfeit websites that look same as authentic ones.

Furthermore, it highlights that phishing has changed dramatically since it was first introduced in the 1990s and become more advanced and prevalent over time. The paper discusses the design of a typical phishing attack that involves three major phases: lure, hook, and catch. The lure is the sending of deceptive messages to attract victims, the hook is directing users to fraudulent websites that look like legitimate websites and the catch is the collection and utilization of the victim's sensitive information.

The discussion also covers different phishing techniques such as spear phishing, clone phishing, malware-based phishing and search engine phishing. The paper also explains how attackers use psychological manipulation techniques such as fear, curiosity and empathy to influence the user behavior and increase the probability of successful attacks. The paper also discusses various detection techniques such as human detection and machine detection. Human detection is where the users are trained and educated while machine detection is where algorithms and systems are used to detect the phishing emails and malicious websites. It has been shown that advanced detection systems such as machine learning models are effective in that they can accurately identify the phishing attacks.

The paper also reviews different countermeasures against phishing such as user training, email filtering, antivirus tools, firewalls, secure communication protocols and multi-factor authentication. Phishing attacks cannot be defended against with one solution, but a multi-layered approach is required. The study found that phishing is a serious and evolving threat and that effective prevention requires a combination of user education, technical measures and organisational policies to mitigate the risk of cyberattacks.

[23] This work reviews the role of machine learning techniques for the detection and classification of phishing attacks highlighting their importance in modern cyber security systems. Phishing is defined as a malicious technique in which attackers imitate trusted websites or services to steal sensitive user information such as login credentials and financial information. The difficulties of phishing detection are acknowledged mainly due to its dependence on human behavior rather than system vulnerabilities. The paper demonstrates that phishing detection can be treated as a classification problem, classifying websites or emails as either legitimate or phishing and it employs several features such as URL-based, domain-based, page-based and content-based features. These features are instrumental in recognizing suspicious websites and malicious activities.

Furthermore, the research provides a comprehensive literature review of various machine learning algorithms used for phishing detection like artificial neural networks (ANN), decision trees (DT), k-means clustering, naïve bayes (NB), random forest (RF) and support vector machines (SVM). Each of the algorithms has been discussed with respect to its working mechanism, advantages, and limitations with some methods achieving high accuracy rates in the detection of phishing attacks. The research reveals that the machine learning techniques are useful in improving the accuracy of the detection of phishing by analyzing patterns from large datasets. The study also mentions that no algorithm is the best in all cases, as the performance depends on selected features, size of the dataset, and the complexity of the problem.

Moreover, the study compares these machine learning methods and provides their advantages and disadvantages. For example, Random Forest is robust and accurate, while Naïve Bayes is simple and fast to compute. On the other hand, Support Vector Machines are accurate but may require careful parameter tuning. The study highlights the importance of machine learning in phishing attack detection and claims that the choice of method depends on the problem and dataset. It recommends the combination of different techniques and features to improve detection effectiveness and enhance cybersecurity systems.

[24] This paper gives a comprehensive survey of intelligent detection methods for phishing attacks with a particular focus on HTML and URL-based techniques. Phishing is defined as a cybercrime that uses social engineering to trick users into giving away sensitive information, such as login credentials and financial data, by means of deceptive messages and malicious links. The paper notes the rapid rise of phishing attacks in recent years and stresses the challenge of detecting them because of how similar they look to legitimate websites. Attackers employ a range of strategies, including misspelled URLs, bogus domains, and shortened links, to ensnare users, thus complicating the detection process even for professionals. The paper reviews several detection methods, among them blacklist-based approaches, traditional machine learning, and deep learning techniques.

Although blacklist methods are straightforward, they fail to identify new phishing URLs. Traditional machine learning requires manual feature extraction which is time consuming and less adaptive. In contrast, deep learning models can automatically extract features and achieve higher accuracy but require large datasets and higher computation resources. Moreover, the study classifies phishing detection techniques based on data types such as URL-based, content-based and hybrid-based methods. URL-based methods analyze the structure of links while content-based methods examine webpage elements like HTML, text and scripts. Hybrid approaches combine both to improve detection performance. Furthermore, the study offers an extensive analysis of deep learning techniques, including data pre-processing approaches like cleaning, tokenization, and feature representation.

It discusses the application of modern algorithms like CNN, RNN, and LSTM to find patterns in phishing data and enhance classification accuracy. The article also reviews popular datasets like PhishTank, OpenPhish, and Alexa that are crucial for training and testing phishing detection models. It also compares different models based on pre-processing techniques, efficiency, feature extraction, and The research shows that deep learning techniques greatly improve the performance of phishing detection systems, especially when used with hybrid features. Still, there are challenges in managing new attack patterns, the complexity of data pre-processing, and the need for large-scale datasets. Future research should focus on improving the model's efficiency and introducing more adaptive detection techniques.

[25] This study is about children's responses to phishing attacks and the impact of security awareness on their ability to detect phishing attacks. Phishing is a major threat to cybersecurity and it is the electronic deception through emails, websites and social media messages to deceive users into disclosing their sensitive information, such as personal information and financial information. Children are especially vulnerable to phishing attacks as they lack the understanding of the principles of online security and privacy. The target population of this study is children between the ages of 7-13 who are increasingly involved in online activities such as gaming, communication and learning. The study also highlights the dramatic increase in children's internet use, the resulting increase in their exposure to online risks, and the need to raise cybersecurity awareness among young users. Two real-world experiments were performed using the WhatsApp application to assess children's behavior.

In the first experiment, children received phishing messages with malicious links with a control group receiving legitimate messages. The results showed that some of the children clicked on the links of the phishing messages, showing that they are vulnerable to such attacks. In the second experiment, participants were presented with another phishing message after receiving awareness training. The results showed that the number of participants who clicked on malicious links was significantly reduced, indicating that security awareness improves children's ability to identify phishing attacks. They also looked at how participants' behavior differed, with some saying they got suspicious messages or blocked the sender, while others didn't even know they were at risk.

Results also show that there was no significant difference in terms of phishing awareness in terms of gender or age. The study concludes with highlighting the importance of security awareness in protecting children from phishing attacks, and observes the need of continued, age-appropriate awareness education for children to improve its effectiveness. The study recommends developing educational programs and materials specifically for children to increase their awareness regarding cybersecurity and reduce their susceptibility to cyber threats.

III. GAP ANALYSIS

A. *Overemphasis on Email Phishing Instead of Social Media Phishing*

Most existing phishing research focuses on email phishing detection techniques, datasets, and prevention strategies. However, social media phishing differs significantly because attackers exploit platform-specific features specific features such as likes and shares [9], [11], [15].

B. *Outdated Detection Models and Limited Datasets*

Many phishing detection systems still rely on the following:

- Signature-based detection.
- Static rule-based approaches.
- Traditional machine learning models trained on outdated datasets [1], [23].

These methods were originally designed for email environments and not effectively to the fast-changing nature of social media platforms.

C. *Lack of Real-Time Detection and Cross-Platform Protection*

The amount of information that social media users share is huge and needs real-time detection and protection, unlike email [21].

D. *Insufficient User Awareness and Behavioral Training*

Human behavior remains one of the biggest vulnerabilities in phishing attacks [13], [16].

E. Emerging Threats from Artificial Intelligence

Artificial intelligence has introduced new phishing techniques, including:

- AI-generated phishing emails and messages.
- Deepfake voice and video impersonation.
- Automated chatbot-based scams.
- Personalized spear-phishing attacks [6], [7].

TABLE I. PREVIOUS STUDIES GAPS

Research Gap	Current Focus in Existing Studies	Weaknesses in Current Solutions	Impact of the Gap	Suggested Future Direction	Ref
Overemphasis on Email Phishing	Most studies focus on email phishing detection and prevention rather than social media phishing.	Existing models fail to analyze social trust, fake profiles, likes, comments, and platform-specific interactions.	Low detection accuracy for phishing attacks on Instagram, Facebook, LinkedIn, and X.	Develop social media-specific phishing detection frameworks based on user behavior and platform interactions.	[1], [9], [11], [15], [20]
Outdated Detection Models	Traditional machine learning and signature-based systems are still widely used.	Cannot detect AI-generated phishing attacks or adaptive phishing techniques.	Increased success rate of modern phishing attacks.	Use deep learning and AI-based adaptive detection systems trained on modern datasets.	[1], [5], [7], [23], [24]
Limited and Outdated Datasets	Existing datasets mainly contain email phishing samples and static URLs.	Lack of real-world social media phishing data and multimedia attacks.	Detection systems perform poorly in realistic environments.	Create updated datasets containing social media phishing scenarios, fake profiles, and AI-generated attacks.	[5], [14], [23]
Lack of Real-Time Detection	Most systems analyze phishing attacks after they occur.	Slow response time and inability to stop phishing propagation quickly.	Rapid spread of phishing attacks through reposts, comments, and messages.	Develop real-time AI-driven monitoring and automated mitigation systems.	[5], [21], [22]
Lack of Cross-Platform Protection	Existing solutions focus on single communication channels only.	Systems cannot detect hybrid phishing attacks across email, SMS, and social media simultaneously.	Attackers exploit multiple platforms to increase phishing success rates.	Design integrated multi-platform phishing detection architectures.	[5], [20], [21]
Weak User Awareness Programs	Awareness programs provide general phishing education.	Training is not personalized and ignores platform-specific behaviors.	Users remain vulnerable to modern social engineering techniques.	Develop adaptive and personalized awareness programs for different user groups.	[10], [13], [16], [17]
AI-Generated Phishing Threats	Limited research addresses AI-generated phishing and deepfake attacks.	Existing systems struggle to identify synthetic text, voice, and video impersonation.	Higher phishing success rates due to realistic AI-generated content.	Develop AI-powered defenses capable of detecting deepfake and generative AI attacks.	[6], [7], [20]
Lack of Real-World Deployment	Many proposed models are tested only in laboratory environments.	Systems are not validated on real social media platforms.	Limited practical applicability of existing research.	Conduct real-world experiments and large-scale deployment testing.	[1], [23], [24]
Limited Protection for Vulnerable Groups	Few studies focus on children, elderly users, and young adults.	Awareness systems do not adapt to age-related vulnerabilities.	Vulnerable groups remain highly susceptible to phishing attacks.	Create demographic-specific phishing awareness and protection mechanisms.	[9], [13], [25]

IV. PROPOSED FRAMEWORK

Based on the weaknesses highlighted across the evaluated literature, this section proposes a theoretical

framework for detecting phishing attempts on social media sites. The framework makes results from the reviewed studies to propose a modular five-stage detection pipeline that overcomes the challenges of single-method approaches, considers human behavioral aspects, and converts technical detection into actionable guidance for users and platform administrators.

A. Framework Overview

This approach is primarily motivated by continuous shortcomings of isolated detection methods that have been reported in the available research. Khan et al. (2025) showcase the blind spots of separate use of NLP, URL analysis, and behavioral analysis; whereas Vanna et al. (2025) validate the superior performance of hybrid models over standalone classifiers and reach detection accuracy levels of 97%-99.82%. In particular, Bossetta (2018) and Silic and Back (2016) demonstrate that traditional platform do not have inbuilt capabilities to identify the mix of fake identities, trust abuse and harmful linking that defines social media phishing. Figure 1 shows the proposed system which addresses these findings by providing a single cohesive pipeline for data collection, feature engineering, detection logic, performance evaluation and user-facing response.

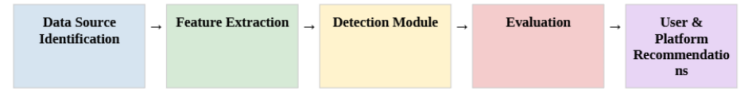


Fig. 1. Proposed Framework Workflow for Phishing Detection on Social Media Platforms

B. Framework Components

- Stage 1: Data Source identification:** The framework begins with identifying the given input data channels to the detection system. Social media phishing is by nature multi-modal: hackers use text messages, anonymous posts, shortened URLs, and account profiles at the same time (Bossetta, 2018; Silic and Back, 2016). Thus, limiting a detection to a single data channel leaves entire attack surfaces uncovered (a shortcoming addressed by Parker and Flowerday, 2020). The framework identifies three main data sources, which include Direct Messages (DMs), that Klutsch et al. (2025) identify as the primary phishing vector on Instagram; shortened or obfuscated URLs, which Asiri et al. (2023) flag as a leading indicator and which traditional blacklists consistently fail to catch given their short average lifespan; and user profile metadata, including account age, follower-to-following ratio, and posting frequency, whose importance both Bossetta's (2018) five-phase attack model and Silic and Back's (2016) fake-profile experiment confirm.
- Stage 2: Feature Extraction:** When the data is collected, the algorithm finds selective traits and classifies them in four categories. A linguistic cues such as "critical," "verify," and "press here" are often identified as phishing

signs by Al-Otaibi and Alsuwat (2020), Bhavsar et al. (2018), and Adu-Manu et al. (2023). Importantly, Madlenak and Hubocan (2025) noted that recent AI-generated phishing attacks no longer have any grammatical flaws, therefore algorithms must advance to evaluating typical patterns in syntax relevant texts. The Uniform Resource Locator (URL) properties such as shortened URLs, IP address in the link, and muddled anchor text, coupled with linguistic characteristics, give high-quality classifications, as confirmed by Asiri et al. (2023) and Jupin et al. (2019). Account behavior features such as posting inconsistencies and juvenile profiles are derived from the research of Alam and El-Khatib.

- c) *Stage 3: Detection Module:* The detection classifier classifies the incoming data into either the phishing or authentic category based on the obtained feature vector. The framework provides two mutually compatible solutions, consistent with the hybrid-model approach of Khan et al. (2025) and Vanna et al. (2025). Rule-based detection employs predetermined heuristic rules (e.g., identifying any communication, including a shortened URL containing an urgency-based keyword) and is characterized by Al-Otaibi and Alsuwat (2020) as interpretable and computationally inexpensive, which makes it appropriate for real-time detecting application. Machine learning classification learns the decision boundaries via labelled samples using algorithms such as Naive Bayes and Logistic Regression. Jupin et al. (2019) claim that Naive Bayes provides a fast computation for high dimensional text classification problems. Vanna et al. (2025) state that supervised classifiers reach accuracy of 90% to 99.45% on phishing datasets. The best approach is the hybrid one: rule-based filters for the obvious cases, while ML classifiers are applied for ambiguous or new type of attacks, including the AI-generated content that Madlenak and Hubocan (2025) identified as a growing blind spot for purely heuristic systems.
- d) *Stage 4: Evaluation:* The performance of the detection module during the evaluation step is evaluated using 3 standardized metrics according to the benchmarks provided in the literature studied. The expected accuracy is in the range of 97%-99.82% as reported by Vanna et al. (2025) for the hybrid mobile phishing detectors. Precision and recall are both important: The need to minimize the fake negatives is emphasized by Chaudhry et al. (2016) and Asiri et al. (2023), since neglected attacks are far more costly than false positives. It must also minimize the false positive rates.

As Parker and Flowerday (2020) write, “habitual users of social media tend to ignore warnings over time, and so sustained precision is needed to maintain user-facing alert credibility.”

- e) *Stage 5: User and Platform Recommendations:* This final step translates detection outcomes into responses targeted at two stakeholder groups, thereby meeting a specific demand mentioned by Onwuzuruoha and Akinola (2024) and Badescu (2024); namely, solutions are commonly suggested with no corresponding user instructions and demographic-specific targeting. In relation to end users, it is advised to include real-time warnings along with explanations of detected risk markers and guidelines for verification of the sender, taking into account Klutsch et al.’s (2025) finding that social media content tends to be processed through heuristics rather than systematic reasoning. As for targeted awareness campaigns, there have been studies which showed positive outcomes in terms of increasing the ability to detect phishing attempts among certain groups: Alwanain (2020) demonstrated significant gains in this regard after brief training among elderly participants; Alwanain (2021) obtained similar findings in children aged between 7 and 13 years; and Yasin et al. (2024) discovered that training via a computer game increased accuracy of recognizing phishing URLs from 27.8% to 80%. Finally, platform administrators are advised to package the detection module as an open-back-end solution, to automatically flag users who exhibit Bossetta’s (2018) ‘Contagion-phase’ behavior.

C. Addressing Literature Gaps Through the Framework

The approach is aimed to immediately address five major shortcomings found in the evaluated literature. First, Khan et al. (2025) and Vanna et al. (2025) find that every one-method detector has different blind spots. To mitigate this problem, the framework fuses the lexical, URL-structural, account-behavioral, and relations features in Stage 2 and forwards it to a hybrid detector in Stage 3. This is implemented in the code by running a rule-based filter and a Naive Bayes classifier in parallel, so that no attack can be missed by either method solely. Second, as shown by Madlenak and Hubocan (2025), AI-generated phishing text is now nearly indistinguishable from legitimate messages. As a result, the ML component in Stage 3 learns distributional content patterns rather than relying on surface errors, correctly flagging even grammatically fluent messages in the code. Third, Parker and Flowerday (2020) and Klutsch et al. (2025) show that follower trust and process heuristics lead to susceptibility, but these are missed by current detectors; the framework incorporates sender relationship as an explicit feature in Stage 2, with contextualized alerts in Stage 5 explaining the related risk. Fourth, Bossetta (2018) and Silic

and Back (2016) found that platforms have no mechanisms to detect bogus accounts before contact is established. The framework addresses this challenge by considering user profile metadata as a first-class data source in Stage 1, enabling the early identification of accounts in the reconnaissance and construction phases of an attack. Fifth, while Alwanain (2020, 2021) and Yasin et al. (2024) show that elderly users, children, and young adults are at disproportionate risk, none of the reviewed frameworks provide demographic-aware guidance. Stage 5 provides this, with tailored awareness recommendations based on empirically documented vulnerabilities for each group. The plain-language alert output of the code is made interpretable regardless of user’s technical background.

D. Illustrative Code Example

The following Python code gives us an example of the Detection Module that we talked about in Stage 3. This example uses a mix of rules and Naive Bayes to look at some social media messages. Each comment that we marked is connected to one of the five gaps, in the literature that we discussed in Section 3.3. The Detection Module is what this code is trying to show. This code is not meant to be used in an application. It is meant to show how the ideas we talked about can be turned into code that can be run. The Detection Module is the thing here.

```

1 # Illustrative Example: Rule-Based + Naive Bayes Hybrid Phishing Detector
2 # Addresses: single-method limitation + AI-generated content detection
3
4 import re
5 from sklearn.naive_bayes import MultinomialNB
6 from sklearn.feature_extraction.text import CountVectorizer
7
8
9 messages = [
10     "Urgent! Verify your account now: http://bit.ly/fake123",
11     "Hey, loved your post! See you at the event tomorrow.",
12     "Your password has expired. Click here to reset immediately.",
13     "Thanks for connecting! Looking forward to collaborating.",
14 ]
15
16 labels = [1, 0, 1, 0]
17
18
19
20 SUSPICIOUS_KEYWORDS = [
21     "urgent",
22     "verify",
23     "password",
24     "click here",
25     "expired"
26 ]
27
28 SHORTENED_URL = r"https?://(bit\.ly|tinyurl|t\.co|goo\.gl)/"
29
30 def rule_based_flag(message):
31     msg_lower = message.lower()
32
33     keyword_hit = any(keyword in msg_lower for keyword in SUSPICIOUS_KEYWORDS)
34     url_hit = bool(re.search(SHORTENED_URL, message))
35
36     return int(keyword_hit or url_hit)
37
38 rule_preds = [rule_based_flag(msg) for msg in messages]
39 print("Rule-Based:", rule_preds)
40
41
42 vectorizer = CountVectorizer()
43 X = vectorizer.fit_transform(messages)
44
45 clf = MultinomialNB()
46 clf.fit(X, labels)
47
48 ml_preds = clf.predict(X)
49 print("Naive Bayes:", list(ml_preds))
50
51 final_preds = [max(r, m) for r, m in zip(rule_preds, ml_preds)]
52 print("Hybrid:", final_preds)
53
54
55 for msg, label in zip(messages, final_preds):
56     if label == 1:
57         print(f"[ALERT] Suspicious message detected: {msg[:60]}...")

```

Fig. 2. Illustrative Python implementation of the hybrid detection module, with annotations linking each component to the identified literature gaps

```

1 Rule-Based: [1, 0, 1, 0]
2 Naive Bayes: [1, 0, 1, 0]
3 Hybrid:      [1, 0, 1, 0]
4
5 [ALERT] Suspicious message detected: Urgent! Verify your account now: http://bit.ly/fa...
6 [ALERT] Suspicious message detected: Your password has expired. Click here to reset imm...

```

Fig. 3. The output if the code

In the output of its code, we can see that among all four messages processed using the hybrid approach: both phishing (malicious) messages are correctly flagged regardless of whether they trigger a validation flag in either or both rule-based samples and ML classifiers, while both legitimate (benign) messages pass through without generating alerts. The effectiveness of Stage 5 can be summarized in the final alert output itself, which serves as a user-facing component: For every flagged message, a plain-language warning is generated that speaks to Gap 5 by providing outputs that users need no programming background to be able to review.

E. The Scope and Limitations

The framework described here is theoretical. “Experimental validation” No empirical implementation or experimental validation has been done as a part of this study. The performance benchmarks cited in Stage 4 are from previous literature and not original experimentation. I must acknowledge three limitations. First, the framework does not consider adversarial adaptation: as Madlenak and Hubocan and Paper 2 (2026) show, AI-generated phishing content evolves fast, and a static classifier will have to be retrained regularly to be effective. Secondly, the relational features suggested in Stage 2 are reliant on the platform graph data that can be restricted by the API policy or data privacy regulations like GDPR and CCPA, as mentioned by Vanna et al. (2025). Third, the framework does not model the game-theoretic interplay between attackers and defenders studied by Aung et al. . Their results display that detection countermeasures can change attacker strategies in a way that calls for adaptive rather than static defence architectures. Future work should focus on overcoming the limitations by empirical evaluation on real world social media datasets, longitudinal evaluation in adversarial settings and incorporating privacy-preserving techniques such as federated learning.

V. GENERAL CONTRIBUTIONS

This study contributes to the body of knowledge in the field of cybersecurity by addressing the growing incidence of social media phishing attacks. Although many previous studies focused mainly on traditional email phishing

attacks, fewer studies focused specifically on phishing techniques in social media environments [11], [12], [21].

Social media phishing differs from traditional phishing because it exploits platform-specific features such as fake profiles, direct messages, likes, comments, and user trust relationships to manipulate victims [9], [11], [15].

Existing phishing detection solutions also face several limitations, including the inability to detect newly generated malicious links, AI-generated phishing messages, highly personalized social engineering attacks, and rapidly evolving phishing strategies [2], [7], [24].

Based on the identified research gaps, this study proposes a framework to address these limitations. The proposed framework provides a more comprehensive approach to phishing detection and prevention on social media platforms. The framework combines URL analysis, behavioral analysis, feature identification, and machine learning-based detection techniques to improve the identification of phishing attempts [5], [17], [23]. Furthermore, the framework emphasizes the importance of combining technical protection mechanisms with user awareness strategies and platform-level recommendations to enhance cybersecurity defenses against modern phishing attacks [10], [18], [22].

The scientific contribution of this work lies in presenting a conceptual framework that combines both technical and human-centered approaches to phishing prevention. Unlike previous studies that focused only on single detection techniques, this framework provides a broader perspective by integrating machine learning concepts, social engineering awareness, phishing detection strategies, and adaptive protection mechanisms within one unified model [6], [20], [23]. In addition, the framework supports future research directions related to intelligent phishing detection systems, AI-assisted cybersecurity solutions, and real-time social media protection systems [5], [24].

From a practical perspective, the proposed framework may help improve cybersecurity practices on social media platforms by supporting the development of more effective phishing detection and prevention tools. The framework may also help reduce risks such as credential theft, identity compromise, malicious link interaction, and social engineering exploitation [8], [19], [21]. Moreover, the proposed model can serve as a foundation for future adaptive and real-time phishing prevention systems capable of addressing evolving cyber threats across multiple social media platforms [5], [6].

Overall, this study contributes to expanding the understanding of phishing attacks in modern social media environments and highlights the importance of integrating technological solutions, machine learning techniques, user awareness, and behavioral analysis to effectively combat evolving cyber threats [2], [17], [20].

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